

2019 Mathematics Kangaroo Contest

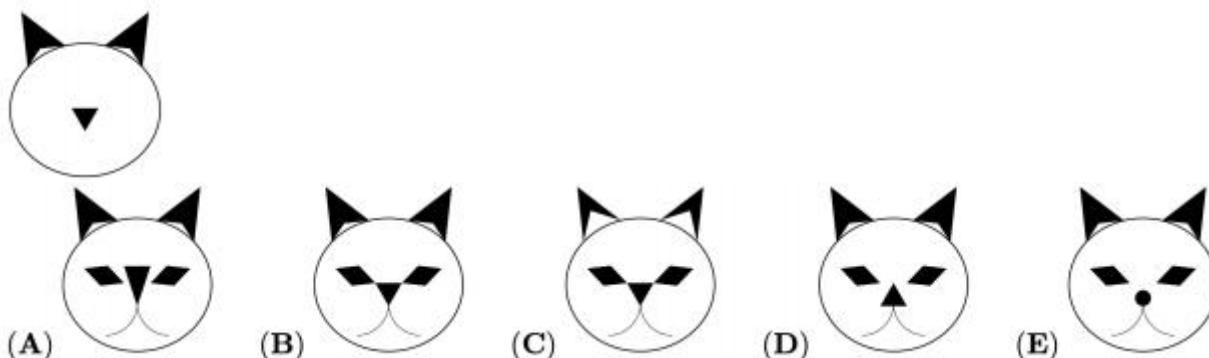
—Benjamin—

2019 数学袋鼠竞赛

Part 1: 10 problems, 3 points each | 第一部分: 10 道题目, 每题 3 分

1 • Carrie has started to draw a cat. She finishes her drawing by adding more color. Which of the figures below can be her drawing?

Carrie 开始画一只猫。她通过添加更多颜色来完成绘图。问下面的哪个图形可能是她的图画?



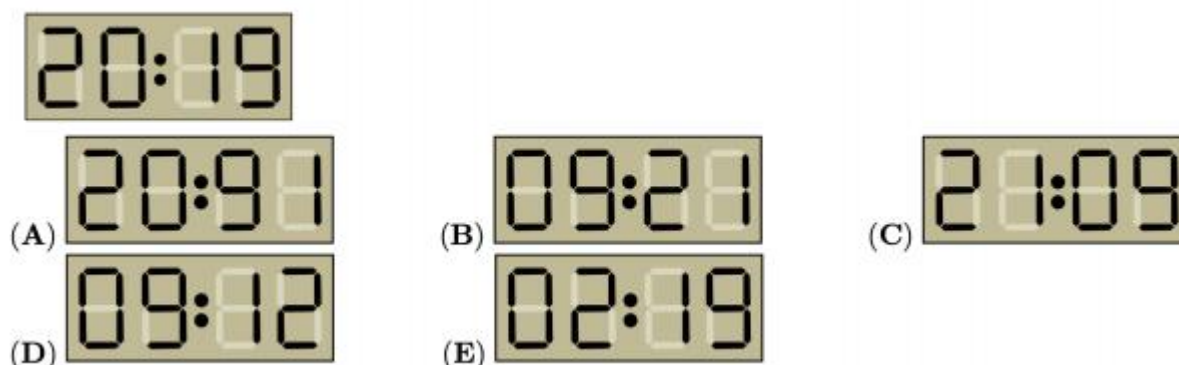
2 • The Mayan people wrote numbers with dots and bars. A dot is written for 1 and a bar for 5. How did they write 17?

玛雅人用点和条来表示数。一个点代表 1，一个长条代表 5。问他们应该怎么写 17?



3 • A digital clock shows the time 20:19. What will the clock show the next time it uses the same digits?

数字时钟显示时间 20:19。时钟在下次使用同样的这些数字时会显示几点?



4 • There are 14 girls and 12 boys in a kindergarten. If half of the children go for a walk, at least how many of them are girls?

幼儿园里有 14 个女孩和 12 个男孩。如果有一半的孩子去散步，那么其中最少有多少个女孩?

- (A) 5 (B) 4 (C) 3 (D) 2 (E) 1

5 • The sum of the dots on opposite faces of an ordinary dice is equal to 7. Which of the following shows the

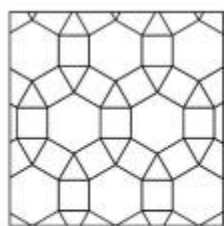
ordinary one?

普通的骰子相对面上的点数之和等于 7。问以下哪一个显示的是普通的骰子？



6 • Which of the following geometric figures is not in this design?

以下哪个几何图形不在这个设计图案中？

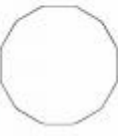


(A)  triangle 三角形

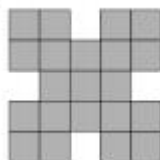
(B)  square 正方形

(C)  regular hexagon 正六边形

(D)  regular octagon 正八边形

(E)  regular dodecagon 正十二边形

7.



Laura wants to colour a 2×2 square  of the above figure. How many possibilities are there?

Laura 想要给上述图形中的一个 2×2 正方形  涂色。共有多少种可能性？

(A)5

(B)6

(C)7

(D)8

(E)9

8 • The 6 smallest odd natural numbers are written on the faces of a dice. Toni throws it three times and adds the results. Which of the following numbers cannot be the sum?

将最小的 6 个奇自然数写在一个骰子的各个面上。Toni 抛掷了这个骰子三次，并且将所得的结果相加。以下哪个数不可能是这个和？

- (A) 21 (B) 3 (C) 20 (D) 19 (E) 29

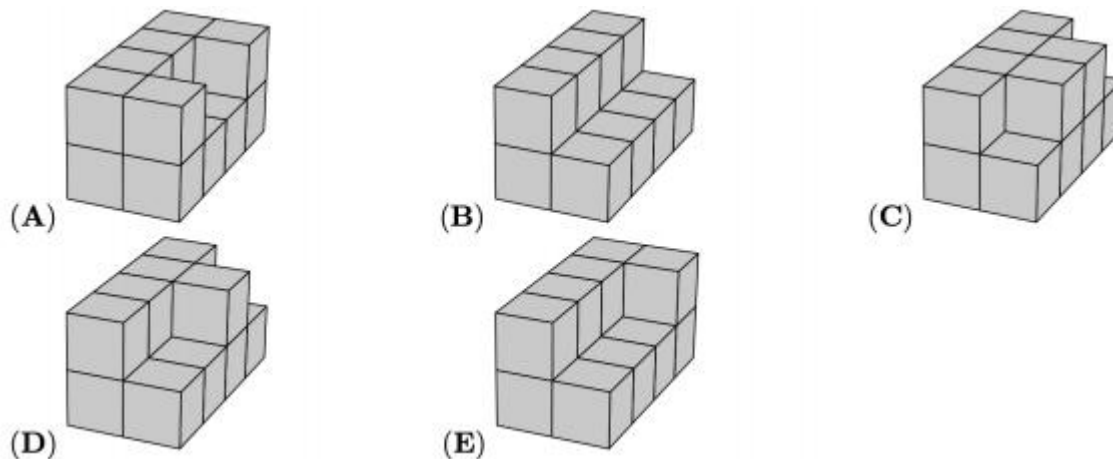
9 • The sum of the ages of a group of kangaroos is 36 years. In two years time the sum of their ages will be 60 years. How many kangaroos are in that group?

一群袋鼠的年龄总和为 36 岁。两年以后，他们的年龄总和将变成 60 岁。问这群袋鼠共有多少只？

- (A) 10 (B) 12 (C) 15 (D) 20 (E) 24

10. Michael paints the following buildings made up of identical cubes. Their bases are made of 8 cubes. Which building needs the most paint?

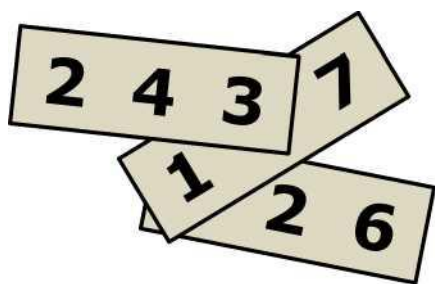
Michael 给由相同立方体组成的以下构件涂油漆。它们的底座都是由 8 个立方体组成。问哪个构件需要最多的油漆？



Part 2: 10 problems, 4 points each | 第二部分：10 道题目,每题 4 分

11. On each of three pieces of paper a three digit number is written. Two of the digits are covered. The sum of the three numbers is 826. What is the sum of the two covered digits?

三张纸片每一张纸上写有一个三位数。其中有两个数字被遮住了。这三个数的总和是 826。问这两个被遮住的数字的和是多少？



(A) 7

(B) 8

(C) 9

(D) 10

(E) 11

12. Riri the frog usually eats 5 spiders a day. When Riri is very hungry, she eats 10 spiders a day. She ate 60 spiders in 9 days. How many days was she very hungry?

青蛙 Riri 通常每天吃 5 只蜘蛛。当 Riri 在很饥饿的状态时，她每天吃 10 只蜘蛛。她在 9 天吃了 60 只蜘蛛。问她有几天很饥饿？

(A) 1

(B) 2

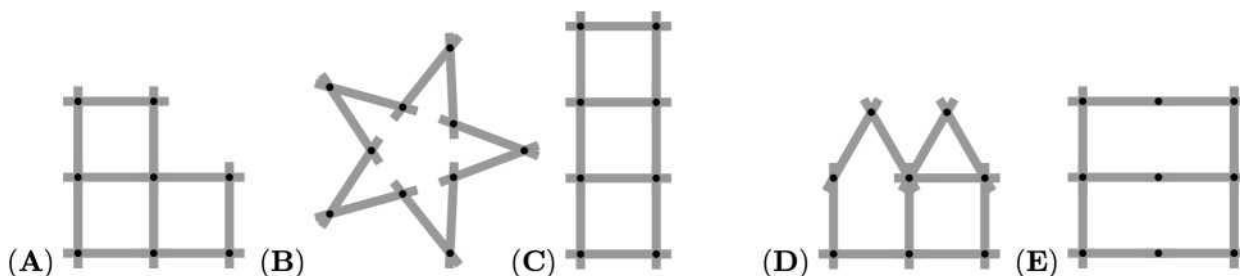
(C) 3

(D) 6

(E) 9

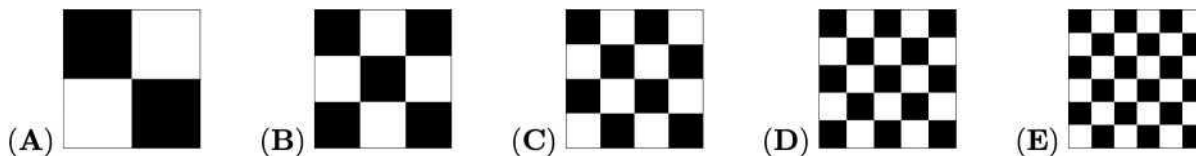
13. Pia plays with a yardstick consisting of 10 sticks (see picture). Which of the following figures cannot be formed with this yardstick?

Pia 在玩一个由 10 节棍子组成的标尺(见图)。下面哪个图形无法用这个标尺拼出？



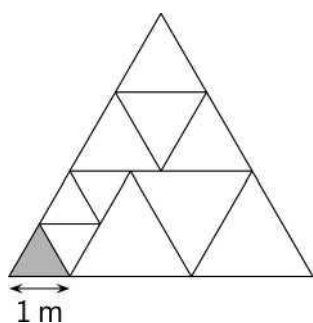
14 • Five equal squares are divided into smaller squares. Which square has the largest black area?

五个相同的正方形被切成了小正方形。问哪个正方形中黑色部分的面积最大？



15 • A big triangle is divided into equilateral triangles as in the figure. The side of the small gray triangle is 1 meter. What is the perimeter of the big triangle in meters?

如图所示，一个大三角形被分割成若干个等边三角形。灰色小三角形的边长为 1 米。问大三角形的周长是多少米？



- (A) 15 (B) 17 (C) 18 (D) 20 (E) 21

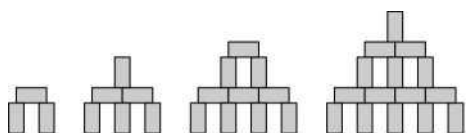
16 • In the garden of a witch there are 30 animals: dogs, cats and mice. The witch turns 6 dogs into cats. Then she turns 5 cats into mice. Now her garden has the same number of dogs, cats and mice. How many cats were there at the beginning?

女巫的花园里有 30 只动物：狗，猫和老鼠。女巫将 6 只狗变成了猫。然后她把 5 只猫变成了老鼠。现在她的花园里有相同数量的狗，猫和老鼠。问开始时有多少只猫？

- (A) 4 (B) 5 (C) 9 (D) 10 (E) 11

17 • With blocks of dimension 1 cm x 1 cm x 2 cm, you can build towers as shown in the picture. How high is a tower that is built in the same way with 28 blocks?

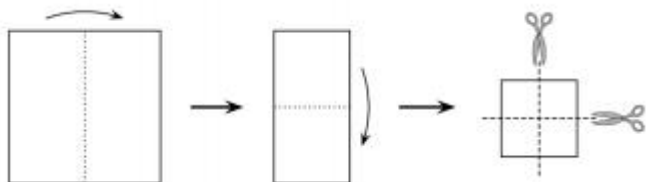
如图所示，可以使用尺寸为 1 cm x 1 cm x 2 cm 的砖来建造塔楼。按同样的方式使用 28 块砖建造的塔楼有多高？



- (A) 9cm (B) 11cm (C) 12cm (D) 14cm (E) 17cm

18 • Bridget folded a square sheet of paper twice, and then cut it twice as shown in the figure. How many pieces of paper will she get?

如图所示，Bridget 将一张正方纸片折叠两次，然后剪了两刀。问她会得到多少片纸？



- (A) 6 (B) 8 (C) 9 (D) 12 (E) 16

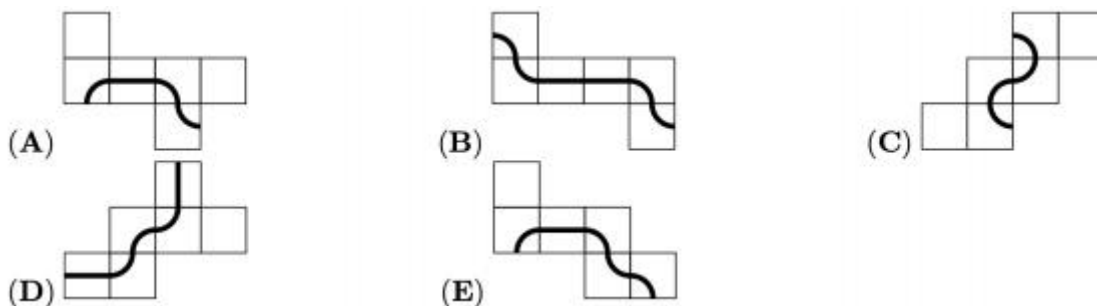
19 • Alex, Bob and Carl go for a walk every day. If Alex doesn't wear a hat, then Bob wears a hat. If Bob doesn't wear a hat, then Carl wears a hat. Today Bob is not wearing a hat. Who is wearing a hat?

Alex, Bob 和 Carl 每天都去散步。如果 Alex 不戴帽子, 那么 Bob 会戴帽子。如果 Bob 不戴帽子, 那么 Carl 会戴帽子。今天 Bob 没有戴帽子。问谁会戴着帽子?

- | | | | |
|--------------------------------------|-------------|----------------------------|-----------------|
| (A) Both Alex and Carl | Alex 和 Carl | (B) Only Alex | 只有 Alex |
| (C) Only Carl | 只有 Carl | (D) Neither Alex, nor Carl | Alex 和 Carl 都没有 |
| (E) It is not possible to determine. | 无法确定 | | |

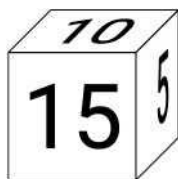
20 • Each of the following pictures shows the net of a cube. Only one of the resulting cubes has a closed line on it. Which one?

以下每张图片都是一个立方体的平面展开图。折成的立方体中只有一个包含有一条闭合的线条。问是哪一个?



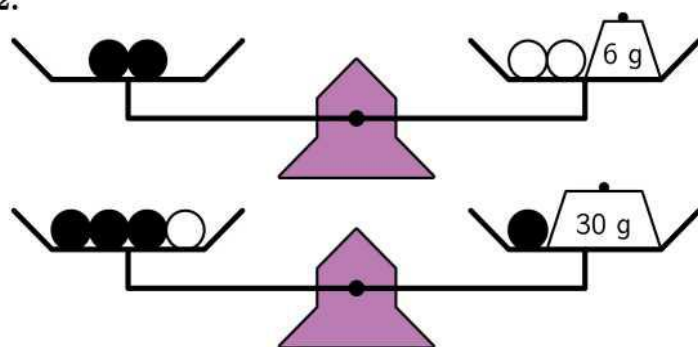
21 • The cube shown in the figure below has a positive integer written on each face. The products of the two numbers on opposite faces are the same. What is the smallest possible sum of the six numbers on the cube?

如下图所示的立方体的每个面上都写有一个正整数。在相对面上的两个数的乘积是相同的。问立方体上六个数的总和最小可能是多少?



- | | | | | |
|--------|--------|--------|--------|--------|
| (A) 36 | (B) 37 | (C) 41 | (D) 44 | (E) 60 |
|--------|--------|--------|--------|--------|

22.



Six identical black beads and three identical white beads are arranged on weighing scales as shown in the picture. What is the total weight of these nine beads?

如图所示，在天平上放置有六颗相同的黑色珠子和三颗相同的白色珠子。问这九颗珠子的总重量是多少？

- (A) 100 g (B) 99 g (C) 96 g (D) 94 g (E) 90 g

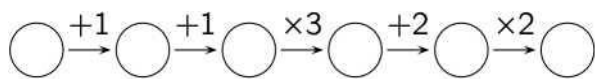
23 • Robert made 5 statements (A) - (E), exactly one of which is false. Which one?

Robert 做出了(A) -(E)的 5 个陈述，其中一个是假的。问是哪一个？

- (A) My son Basil has 3 sisters. 我的儿子 Basil 有三个姐妹
 (B) My daughter Ann has 2 brothers. 我的女儿 Ann 有两个兄弟
 (C) My daughter Ann has 2 sisters. 我的女儿 Ann 有两个姐妹
 (D) My son Basil has 2 brothers. 我的儿子 Basil 有两个兄弟
 (E) I have 5 children. 我有 5 个孩子

24 • Benjamin writes an integer in the first circle and then fills the other five circles by following the instructions. How many of the six numbers in the circles are divisible by 3?

Benjamin 在第一个圆圈中写入一个整数，然后按照指示在其他五个圆圈中填数。问圆圈中的六个数有多少个能被 3 整除？

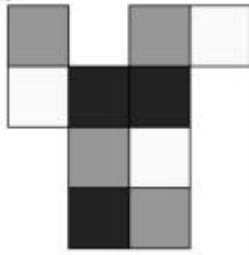


- (A) 1 1 个
 (B) both 1 and 2 are possible 可能是 1 个或 2 个
 (C) 2 2 个
 (D) both 2 and 3 are possible 可能是 2 个或者 3 个

(E) both 3 and 4 are possible

可能是 3 个或者 4 个

25.



The cardboard is folded into a $2 \times 1 \times 1$ box. Which picture does NOT show this box?

纸板折叠成一个 $2 \times 1 \times 1$ 的盒子。下面哪张图片显示的不是这个盒子？



26 • Emily took selfies with her 8 cousins. Each of the 8 cousins is in two or three pictures. In each picture there are exactly 5 cousins. How many selfies did Emily take?

Emily 和她的八个表兄弟一起拍了自拍照。8 个表兄弟中的每一个都在两张或三张照片中。而每张照片中都恰好有 5 个表兄弟。问 Emily 拍了多少张自拍照？

(A) 3

(B) 4

(C) 5

(D) 6

(E) 7

27 • Jette and Willi are throwing balls at two identical pyramids of 15 cans. Jette knocks down 6 cans with a total of 25 points. Willi knocks down 4 cans. How many points does Willi score?

Jette 和 Willi 向两个完全一样的由 15 个罐头组成的金字塔掷球。Jette 击倒了 6 个罐头并得到 25 分。Willi 击倒了 4 个罐头。问 Willi 会得多少分？



(A) 22

(B) 23

(C) 25

(D) 26

(E) 28

28. Every digit on my digital clock is composed of at most 7 segments, as follows:

我的数字时钟上的每个数字最多由 7 段组合而成，如下所示：



But, unfortunately, in every set of 7 segments the same 2 segments don't work. At this moment my clock shows :

然而，在每个显示数字的 7 段组中，都有相同的 2 段不起作用。此刻我的时钟显示为



What will it show after 3 hours and 45 minutes?

问 3 小时 45 分钟后会怎样显示？



29 • Linas builds a $4 \times 4 \times 4$ cube using 32 white and 32 black $1 \times 1 \times 1$ cubes. He arranges the cubes so that as much of the surface of his large cube is white. What fraction of the surface of his cube is white?

Linas 使用 32 个白色和 32 个黑色 $1 \times 1 \times 1$ 立方体构建一个 $4 \times 4 \times 4$ 立方体。他排列这些立方体使得大立方体的表面上有尽可能多的是白色。问大立方体的表面中白色占几分之几？

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{2}{3}$

(D) $\frac{3}{4}$

(E) $\frac{3}{8}$

30 • Zev has two machines: one exchanges 1 white token into 4 red tokens, while the other exchanges 1 red token into 3 white ones. Zev has 4 white tokens. After exactly 11 exchanges, he has 31 tokens. How many of those are red?

Zev 有两台机器：一台将 1 个白色代币换成 4 个红色代币，而另一台则将 1 个红色代币换成 3 个白色代币。Zev 有 4 个白色代币。经过恰好 11 次交换，他有 31 个代币。问其中有多少个是红色的？

(A) 21

(B) 17

(C) 14

(D) 27

(E) 11

答案：BCCEE DDCBA CCCBA CBCAD CEDCB BDADC